

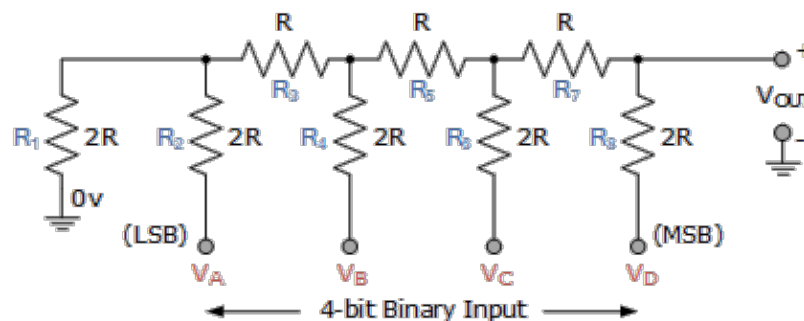
Switching Circuits Laboratory (CS29002)

Assignment 8 (25-03-2026)

- a) Design a modulo-16 **ripple counter** using J-K flip-flops, and trace the output waveforms on the oscilloscope.

Modify the counter to make it a modulo-12 counter.

The following is a resistive-ladder digital-to-analog converter that produces a voltage V_{OUT} proportional to the digital word (V_D, V_C, V_B, V_A) applied as input. Connect such a circuit to the modulo-12 counter, and trace the output waveform on the oscilloscope. Choose $R = 1 \text{ kohm}$.



- b) Design a 3-bit synchronous counter using the minimum number of J-K flip-flops and any required additional gates, which would count in the following sequence: 000, 011, 100, 110, and then back to 000. Assume that the counter starts in the initial state 000. Apply a slow clock and verify the counter's operation by observing the outputs on the on-board LEDs.
- c) [**PAPER DESIGN ONLY**] Modify the design of the counter as designed in part (b) such that if the counter starts with any initial state (other than the four 3-bit states mentioned), it will move to state 000 in the next clock. You need to include this design only in the PDF report, not implement it in the lab.

Instructions:

- 1) For part (b), you need to show the complete design to the TA/faculty, and if they approve, you can proceed to issue the components.
- 2) Prepare a laboratory report in PDF format and upload it to Moodle by the specified deadline. A single report has to be uploaded per group.